13. The dimension of $\left(\mu_0\varepsilon_0\right)^{-\frac{1}{2}}$ is
[Main AIPMT 2011]

(a) length (b) time

(c) velocity (d) force

14. Planck's constant (h), speed of light in vacuum (c) and Newton's gravitational constant (G) are taken as three fundamental constants. Which of the following combinations of these has the dimension of length?.
[NEET 2016 (phase II)]

(a) $\frac{\sqrt{hG}}{c^{\frac{3}{2}}}$ (b) $\frac{\sqrt{hG}}{c^{\frac{5}{2}}}$

(c) $\sqrt{\frac{hc}{G}}$ (d) $\sqrt{\frac{Gc}{h^{\frac{3}{2}}}}$

15. A length-scale (l) depends on the permittivity (ε) of a dielectric material, Boltzmann constant (k_B), the absolute temperature (T), the number per unit volume (n) of certain charged particles, and the charge (q) carried by each of the particles. Which of the following expression for l is dimensionally correct?. [JEE (advanced) 2016]

(a) $l=\sqrt{\frac{nq^2}{\varepsilon k_B T}}$

(b) $l=\sqrt{\frac{\varepsilon k_B T}{nq^2}}$

(c) $l=\sqrt{\frac{q^2}{\varepsilon n^{\frac{2}{3}} k_B T}}$

(d) $l=\sqrt{\frac{q^2}{\varepsilon n k_B T}}$

Answers:

1) a	2) d	3) d	4) c
5) d	6) c	7) b	8) a
9) d	10) b	11) c	12) d
13) c	14) a	15) b	

II. Short Answer Questions

1. Briefly explain the types of physical quantities.
2. How will you measure the diameter of the Moon using parallax method?
3. Write the rules for determining significant figures.
4. What are the limitations of dimensional analysis?
5. Define precision and accuracy. Explain with one example.

III. Long Answer Questions

1. i) Explain the use of screw gauge and vernier caliper in measuring smaller distances.
   ii) Write a note on triangulation method and radar method to measure larger distances.
2. Explain in detail the various types of errors.
3. What do you mean by propagation of errors? Explain the propagation of errors in addition and multiplication.

UNIT-1(XI-Physics_Vol-1).indd 38 28-01-2020 13:37:23